

5.DEVELOPMENT

The development phase turned the design specifications into a functional, live application within the ServiceNow development sandbox environment. Engineering efforts focused on leveraging ServiceNow's native, low-code capabilities to build a modern, sustainable solution that avoids the technical debt often associated with heavy custom scripting.

Front-End Form Design & Data Validation Rules

The user entry point was established by constructing the "Request a Laptop for Work Purposes" Service Catalog item from scratch. The form layout follows clean UI design patterns, arranging fields into logical sections for employee details, equipment specifications, and business justifications. This structured layout guides users smoothly through the submission process, minimizing form completion friction.

To maintain high data quality, strict validation rules were enforced directly at the presentation layer. Catalog UI Policies were implemented to make fields like Employee ID, Laptop Type, and Business Justification mandatory under specific conditions. By applying these validation rules at the client level, the system blocks incomplete submissions immediately, keeping poor data from reaching backend tables and reducing clean-up work for IT support teams.

Backend Automation via Flow Designer

Backend process orchestration was built entirely within Flow Designer, using a clean, graphical flow structure instead of legacy, script-heavy workflow components. A custom application flow was configured to trigger automatically whenever a new request is logged against the laptop catalog item. This flow automatically reads the catalog variables and maps them directly into downstream process activities.

The flow's primary action node extracts the requester's manager profile information and generates an active record within the ServiceNow approval engine. The workflow then enters a wait state, holding further processing until the manager submits an official response. This design naturally supports async operations, keeping platform processing threads light and responsive while requests await review.

Once approved, the flow advances to an operational task creation block, which automatically generates a new record in the Catalog Task table (sc_task). This task is automatically routed to the hardware fulfillment group's queue and populates with all user details, equipment choices, and business justifications needed for setup and delivery. Outbound communication actions were also integrated into the flow, sending automated status emails to employees as requests progress through approval and fulfillment stages.

Completed System Integration Milestones:

- Catalog Form Validation:** Deployed client-side validation rules across all mandatory inputs to ensure high data quality.
- Dynamic Flow Routing:** Built automated manager lookup and approval routing logic within Flow Designer.
- Fulfillment Task Generation:** Configured automatic IT hardware task creation and data population upon request approval.
- Automated Communications:** Set up multi-stage email notification alerts to keep all stakeholders updated automatically.

Comprehensive end-to-end testing verified that all configuration paths function correctly. Test runs confirmed that records pass smoothly through approval and task creation states, validation rules block bad data effectively, and system notifications deploy as expected, demonstrating a reliable, fully automated fulfillment workflow.

Order Status

[Back to Catalog](#)[Continue Shopping](#)[Home](#)

✔ Thank you, your request has been submitted

Order Placed: 2026-06-18 02:33:41

Request Number: [REQ0010007](#) ☆

Estimated Delivery Date of Complete Order: 2026-06-20

Description	Delivery Date	Stage	Price (ea.)	Quantity	Total
Request a laptop for work purposes	2026-06-20	▶✔🔄🕒🕒		1	
				Total	-

[Back to Catalog](#)[Continue Shopping](#)

[Home](#)

🔍

 Laptop Request Process Flow

Active

View:







TRIGGER

 Service Catalog

Trigger

Service Catalog

Advanced Options

Close

 Laptop Request Process Flow

Active



View:





Test

Edit flow

Deactivate

...

1

 Ask For Approval on Requested Item



Action Properties

Action

Ask For Approval

Action Inputs

* Record

Trigger ...

 Requested Item ...

Table

Requested Item [sc_req_item]

Approval Reason

Approval Field

Approval

Journal Field

Approval history

* Rules

Approve

 When:

All users approve

System Administrator X

 OR AND

Due Date

None

Add another OR rule set

Data

Collapse All

Flow Variables

Trigger - Service Catalog

Requested Item Record Record

Run Start Time UTC Date/Time

Table Name Table Name

Run Start Date/Time Date/Time

1 - Ask For Approval

Approval State Choice

Action Status Object

2 - If

3 - Create Record

Catalog Task Record Record

Catalog Task Table Table

Action Status Object

Read-only | Status: Published | Application: Smart Laptop Request Management

2

 If

Condition Label:

* Condition 1:

1 - Ask For Ap...

 Approval St... is

Approved

 or and

Add another condition set(OR)

Close

Close

